

Derbyshire Shared Care Pathology Guidelines

Testosterone deficiency (Primary & Secondary Hypogonadism) in Adult Males

Introduction

Testosterone is essential for the development and maintenance of adult male secondary sexual characteristics. Testosterone deficiency is a syndrome characterised by the occurrence of both clinical (relevant signs & symptoms) and biochemical abnormalities. Its prevalence increases with age and can be found in >2% of men over the age of 40 (See reference 1).

- General population screening for testosterone deficiency is not recommended, but biochemical testing should be performed where clinical signs or symptoms are present
- The most common findings include, but are not limited to: decreased libido, erectile dysfunction, decrease in body hair, fatigue, low mood, gynaecomastia, hot flushes, osteoporosis, small testes, central obesity and sleep disturbances

- An increasing theme from referrals received, is patients who have been diagnosed and had treatment commenced for hypogonadism in the Private sector
- The default position would be that treatment commenced in the Private Sector should remain there
- If patients wish to convert to NHS treatment, then that would only be considered where patients are able to demonstrate that the diagnostic thresholds described in this guideline have been met prior to treatment
- Where this is not the case, testosterone treatment would need to be stopped for 6 months to allow retesting to occur before referral can be accepted
- Patients also need to be aware that they would only be considered for forms of testosterone replacement available in the NHS

Initial investigations

- Clinical history should try to establish relevant signs and symptoms that indicate testosterone deficiency
- If erectile dysfunction is the predominant finding then performing the "Sexual Health Inventory for Men (SHIM)" can be useful in determining the effect of treatment (Reference 2)
- Social history concentrating on factors known to result in a decline in testosterone should be considered e.g. alcohol use, illicit drug abuse, opiate misuse, anabolic steroid abuse and obesity
- Since there is a degree of diurnal variation in hormones (including testosterone), blood samples should always be taken around 9 am (between 8-10am) after sleep (i.e. not after a night shift)

First line investigations (only if signs or symptoms present):

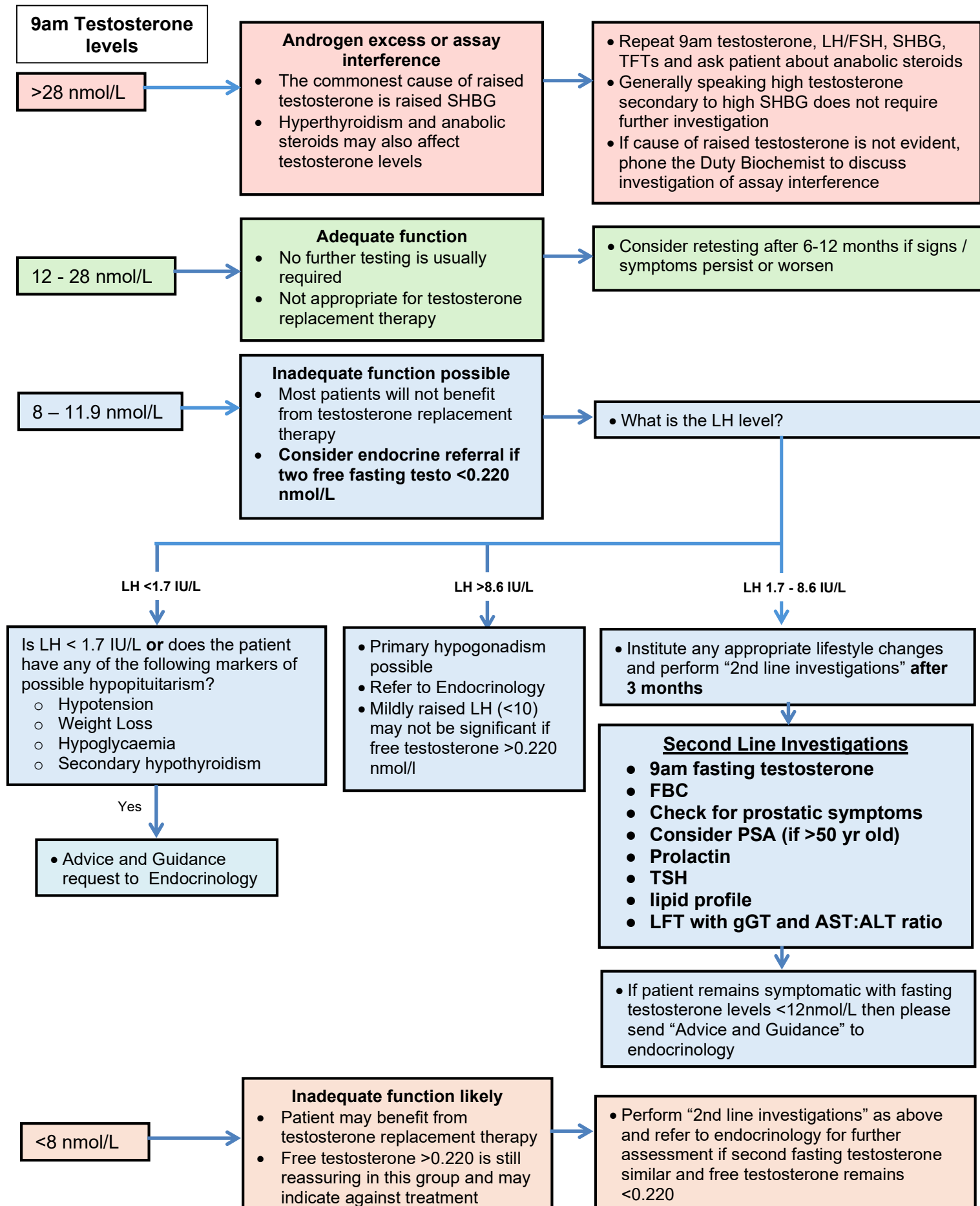
- **9am Testosterone (Ideally fasted)**
- **HbA1c**
- **Blood pressure check**



Unless already measured, the laboratory will add LH and FSH to all samples where the testosterone is <12 nmol/L and will also report the calculated free testosterone level



See page 3 for interpretation of testosterone levels



What to do with a patient with low testosterone plus diabetes?

The primary indication for testosterone replacement therapy remains improving symptoms of hypogonadism. Therefore possible treatment should be discussed with a specialist on that basis (please request 'Advice and Guidance' or refer to Diabetes/Endocrinology). If a trial of testosterone replacement is advised, it is then important to confirm symptomatic benefit. If no benefit is demonstrated then cessation of treatment should be considered.

Testosterone replacement therapy

Testosterone replacement therapy should NOT be initiated in primary care without either requesting 'Advice and Guidance' or after referral to Endocrinology. On-going monitoring can be done in primary care after agreement with secondary care.

Relative Contra-indications to Treatment

- Polycythaemia (unless with agreement from a Consultant Haematologist)
- Raised PSA (unless with agreement from Urology). PSA must be checked before treatment in all men >40yr
- International prostate symptom score >19 (unless with agreement from Urology)
- Testosterone treatment will never be appropriate for patients who are wanting to prioritise fertility. In the subfertility context, early referral to Fertility services may be more useful than Endocrinology review.

Monitoring patients on testosterone

Once established on testosterone replacement therapy by secondary care (or after appropriate agreement through 'Advice & Guidance'), ongoing monitoring can take place in primary care.

Monitoring required and timing of testing

- Check PSA (>40yrs), FBC and Testosterone levels within 6 months after commencing testosterone and then annually thereafter.
- After a change in frequency or dose, repeat the monitoring tests within 6-12 months and seek Endocrinology A&G for further advice
- Timing of testosterone measurement is determined by the preparation of testosterone replacement.
 - Testosterone Gel- post gel 2-5 hours (peak)
 - IM Long acting Testosterone undecanoate- trough within the week before due
 - IM Short acting Sustanon- testosterone measurement is not validated, ensure normal HCT is maintained.

Interpretation of results

- Testosterone levels should be checked to ensure safety that the patient is not over-replaced, however adjustments are usually based on ongoing symptoms.
- If Haematocrit increases to >0.52, please pause testosterone replacement and seek Endocrinology advice through Advice and Guidance.
- A persistently raised or significantly increasing PSA should warrant advice request or referral to Urology

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- Alteration of Testosterone dosing
 - Increase daily gel dose or IM testosterone undecanoate 1000mg frequency only if symptomatic **and** sub-optimal testosterone level (<11nmol/L)
 - Decrease daily gel dose or IM testosterone undecanoate 1000mg frequency if high HCT **or** supra-therapeutic testosterone level on two checks (that fulfil above conditions)
 - When prescribing testosterone gel be aware of the various testosterone concentrations of different products and consult specialist through Advice and Guidance if considering switch of preparation
 - Bone health assessment for patients with hypogonadism is only required every 3 years if other risk factors for fragility fracture or if secondary care have advised this on discharge because of low BMD at baseline. Patients require consideration of additional bone protection whilst on adequate testosterone replacement if a fragility fracture occurs.

Contacts

UHDB

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131
Endocrinology Advice (urgent)	07879 115507 (9am – 5pm, Mon – Fri)
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271
Endocrinology Advice (urgent)	01246 513104
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Patient Information website

<http://www.patient.co.uk/medicine/Testosterone.htm>

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Review history	Version/Date	Authors	Reason for review
	V1 Nov 2011	Dr J Monaghan, Dr Rustam Rea, Dr R Stanworth, Dr P Masters, Dr N Lawson	New Document
	V2 Nov 2014	Dr R Stanworth, Dr P Masters Blackwell, Mrs H Seddon	Routine Review
	V3 Sept 2018	Dr R Stanworth, Blackwell, Mrs H Seddon	Routine Review
	V4 Jul 2021	Dr A Ugur, Dr D Hughes, Dr R Robinson, Dr P Blackwell, Mrs H Seddon	Routine Review
	V5 Mar 2022	Dr A Ugur, Dr D Hughes, Dr R Robinson, Dr P Blackwell, Mrs H Seddon	Minor flow chart update

	V6 Mar 2024	Dr A Ugur, Dr D Hughes, Dr R Robinson, Dr P Blackwell, Mrs H Seddon	Routine Review
Approval	Date	Name(s)	Designation
Secondary Care UHDB	22.04.26	Dr A Ugur, Dr R Stanworth, Dr D Hughes	Consultant Endocrinologists
Secondary Care CRH	24.04.26	Dr R Robinson	Consultant Endocrinologist
Primary Care	27.04.26	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	28.04.26	Ms S Knowles	Consultant Clinical Scientist

References

1. Wu, Frederick C. W., Abdelouahid Tajar, Jennifer M. Beynon, Stephen R. Pye, Alan J. Silman, Joseph D. Finn, Terence W. O'Neill, et al. 2010. "Identification of Late-Onset Hypogonadism in Middle-Aged and Elderly Men." *The New England Journal of Medicine* 363 (2): 123–35.
2. Dawson, Chris, and Gordon Muir. 2005. *The Evidence for Urology*. Tfm Pub Limited.

Physiology

Gonadotrophin releasing hormone (GnRH) is secreted in a pulsatile manner from the hypothalamus. GnRH causes the release of lutenising hormone (LH) and follicle stimulating hormone (FSH) from the anterior pituitary. In the testis, FSH acts on the Sertoli cells, causing maturation of the germ cells to stimulate spermatogenesis. The Sertoli cells also produce Inhibin B which feeds back on the pituitary to reduce FSH production. LH stimulates the secretion of testosterone from the Leydig cells of the testis. Testosterone feeds back on the hypothalamus and pituitary decreasing GnRH and LH production. Peak adult testosterone occurs around 9.00 am, whilst the lowest values (up to 60% lower) are found in the evening.

Testosterone is almost all bound to transport proteins: 60% of plasma testosterone is bound to Sex Hormone Binding Globulin (SHBG), 38% loosely bound to albumin and other binding proteins whilst the rest is "free" or unbound. This free testosterone is considered to be the physiologically active form, and will obviously be dependent on the total amount of SHBG present. Factors that decrease SHBG include androgens, obesity, hyperinsulinism, hypothyroidism and glucocorticoids. Raised SHBG levels can be found in hyperthyroidism. Measurement of free testosterone is technically difficult to achieve and therefore calculations using total testosterone/SHBG are usually used.